

Section 6 Solution

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3 Exercise

3.1 Preferences and Utility Functions

Suppose there are two goods in the world: apples (x) and oranges (y), and that our preferences can be represented by the following utility function: $u(x, y) = 2x + y$

1. If a consumer does not have any apples, how many oranges would he need to be indifferent to a consumption bundle containing 2 apples and 1 orange?

Answer: 5

2. Draw the indifference curves for when the utility equals 5 units and when the utility equals 7 units. Do these indifference curves cross? Can they ever cross? Why or why not?

Answer: The equations describing these curves are: $y = 7 - 2x$ and $y = 5 - 2x$ which are easy to draw. They can't cross, won't cross (different intercept, same slope). If they crossed, that would imply that a bundle of goods can have two different utilities, which makes no sense.

3. Prove that the preferences represented by this utility function are transitive. That is, show that if $(x_1, y_1) \succsim (x_2, y_2)$ and $(x_2, y_2) \succsim (x_3, y_3)$, then $(x_1, y_1) \succsim (x_3, y_3)$.

Answer:

$$2x_1 + y_1 \geq 2x_2 + y_2$$

$$2x_2 + y_2 \geq 2x_3 + y_3$$

by transitivity of $\geq$, it follows that:

$$2x_1 + y_1 \geq 2x_3 + y_3$$

3.2 Utility function

Consider the following utility function, with $\alpha > 0$ and $\beta > 0$:

$$u(x, y) = \alpha \log(x) + \beta \log(y)$$

Recall the definition of **monotonic preferences** (basically, *more is not worse*).

1. Are the preferences represented by the above utility function monotonic? Explain.

Answer: Yes, because the utility function is greater when we have more of each good. That is, our marginal utilities are positive, or $\frac{\partial u(x, y)}{\partial x} = \frac{\alpha}{x} > 0$ and $\frac{\partial u(x, y)}{\partial y} = \frac{\beta}{y} > 0$

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2. What is the sign of the slopes of our indifference curves? Are they upward sloping or downward sloping?

Answer: Using IFT, we know that the slope of our indifference curves, represented on Cartesian coordinates as $\frac{dy}{dx}$, is as follows

$$\frac{dy}{dx} = -\frac{\partial u / \partial x}{\partial u / \partial y} = -\frac{\alpha/x}{\beta/y} < 0$$

Since all 4 of the elements in this expression are positive, the above slope must be negative, so MRS is always positive, or our indifference curves are all downward sloping.

3. Let's consider another utility function, the utility function: $\tilde{u}(x, y) = x^\alpha y^\beta$. Prove that the two utility functions u and $\tilde{u}$ represent the same preferences; that is, prove that

$$u(x_1, y_1) \geq u(x_2, y_2) \Leftrightarrow \tilde{u}(x_1, y_1) \geq \tilde{u}(x_2, y_2)$$

Answer: We know that if we can find a monotonic function $f(\cdot)$ such that $\tilde{u}(x, y) = f(u(x, y))$, then this works. Consider: $f(x) = \exp(x)$, then it's easy to see that $\exp(\alpha \log(x) + \beta \log(y)) = x^\alpha y^\beta$.

4. Suppose currently market price of x is p , price of y is 1. Currently consumer is consuming at the point (x_0, y_0) . What would consumer do if $MRS(x_0, y_0) > p$? What would consumer do if $MRS(x_0, y_0) < p$?

Answer: Think about the intuition of $MRS = \frac{dy}{dx}$ again. It means how much y you're willing to give up for a unit of x . Now if $MRS(x_0, y_0) > p$, then it means you can actually just give up p units of y for a unit of x , which is lower than your willingness to pay. Thus you'll like to buy more x and less y . Likewise, if $MRS(x_0, y_0) < p$, then consumer is willing to buy less x and more y .

Notice that this utility function is concave (i.e. MRS is decreasing in x , see part (2)). So $MRS(x_0, y_0)$ would be converging to p until they are equal. **This is exactly the FOC of utility maximization.**

5. Suppose now the utility function has changed to $u(x, y) = 2x + y$. How would your answer in Part (4) differ?

Answer: The same logic can be applied, that is, consumer would buy more x if $MRS(x_0, y_0) > p$, and vice versa. The difference is that MRS is no longer a decreasing function in x ; MRS is a constant and won't converge to p . Therefore, consumer would keep buying more and more x until she spends all her money on x .

1 A Brief Review

1.1 Utility maximization

Formally, our optimization problem is as follows in the two-good case:

$$\max_{x_1, x_2} u(x_1, x_2) \text{ s.t. } \underbrace{p_1 x_1 + p_2 x_2 \leq M}_{\text{budget constraint}}, \quad x_1 \geq 0, \quad x_2 \geq 0$$

In other words, we want to choose the quantities of goods x_1 and x_2 that maximize our utility, subject to the following constraints:

1. The bundle does not cost more than our income M .
2. We consume non-negative amounts of each good.

Budget Constraint: Check whether the **marginal utility** with respect to x_i , or $\frac{\partial u}{\partial x_i}$ is *strictly* positive for all values of (x_1, x_2) . If so, then preferences are *strictly monotonic* in x_i (meaning *more is better*). In this case, a utility-maximizing individual will not leave any portion of M unspent, because she could always increase her utility by purchasing more of x_i .

After showing $\frac{\partial u}{\partial x_i} \geq 0$ for all x_i and for some j , we have $\frac{\partial u}{\partial x_j} > 0$, then our budget constraint will hold with equality, that is, $p_1x_1 + p_2x_2 = M$. In which case, you write the normal Lagrangian:

$$\mathcal{L}(\lambda, x_1, x_2; p_1, p_2, M) = u(x_1, x_2) - \lambda(p_1x_1 + p_2x_2 - M)$$

Positivity Constraints: We deal with these by trial-and-error. Ignore them when you write the Lagrangian, but once you have solved for x^* , we have to check whether the solution satisfies $x_i \geq 0$ for all i . If it doesn't, then x^* is NOT a solution to the maximization problem and you need to check other possibilities, namely corner solutions (when you spend all of M on one good and 0 on the other).

Solution:

- What's the graphic intuition of FOC?
- Can we have interior solution for convex utility functions? Why?

1.2 Indirect Utility Function

• **Definition:**

The indirect utility function takes prices and income as its arguments and gives you the maximum achievable utility. Here is the two-dimensional case (2 prices, 2 goods):

$$v(p, M) = u(x_1^*(p, M), x_2^*(p, M); p, M)$$

• **Properties:**

1. As we increase M the set of affordable bundles (the budget set) increases. Maximizing a function over a larger set will only increase the value at the optimum. Thus, $\partial v(\mathbf{p}, M)/\partial M \geq 0$.
2. As we increase p_i , the set of affordable bundles (the budget set) decreases. Maximizing a function over a smaller set will only decrease the value at the optimum. Thus, $\partial v(\mathbf{p}, M)/\partial p_i \leq 0$.
3. Note that we can use the Envelope Theorem to find how our (indirect) utility changes with prices and our income. For example:

$$\frac{\partial v(p, M)}{\partial p_1} = \frac{\partial \mathcal{L}(\lambda^*, x_1^*, x_2^*; p_1, p_2, M)}{\partial p_1} = \frac{\partial u(x_1^*, x_2^*)}{\partial p_1} - \lambda^* x_1^*$$

Our prices usually don't appear in the utility function, in which case, the above expression reduces to the following: $\frac{\partial v(p, M)}{\partial p_1} = -\lambda^* x_1^*$.

Likewise, $\frac{\partial v(p, M)}{\partial M} = \lambda^*$. Thus we can interpret λ as marginal utility of an extra dollar of consumption expenditure, or marginal utility of "income".

2 Exercises

2.1 Review of basic utility functions

Derive MRS for the following four utility functions, roughly draw the indifference curves, write down Marshallian demand functions of x and y .

- Cobb-Douglas: $u(x, y) = x^\alpha y^\beta$

- Stone-Geary: $u(x, y) = \alpha \log(x - c) + \beta \log y$
- Perfect substitute: $u(x, y) = \alpha x + \beta y$
- Perfect complement: $u(x, y) = \min(\alpha x, \beta y)$
- CES: $u(x, y) = (\alpha x^\rho + \beta y^\rho)^{\frac{1}{\rho}}$

2.2 Midterm 1, 2008

Yassine likes three goods: consumption goods x_1 , x_2 , and leisure l . He maximizes the utility function $u(x_1, x_2, l) = x_1^{\alpha_1} x_2^{\alpha_2} l^\gamma$ With $0 < \alpha_i < 1$ for $i = 1, 2$ and $0 < \gamma < 1$. The consumption good x_i has price p_i (for $i = 1, 2$), the hourly wage is w .

1. Consider that Yassine has H hours to work and, if he does not work, he takes leisure. For example, H could be 24 hours if the time period is a day. Hence, the hours worked h equals $H - l$. There are no sources of income other than the income from hours worked. Write down the budget constraint as a function of x_1 , x_2 , and l . [Hint: money spent on goods has to be smaller than or equal to money earned.]
2. Write down the maximization problem of the worker with respect to x_1 , x_2 , and l with *all* of the relevant constraints. Assume that the budget constraint is satisfied with equality. Why can we assume this? Provide as complete of an explanation as you can.
3. Write down the Lagrangian and derive the first order conditions with respect to x_1 , x_2 , l and λ .
4. Solve for x_1^* as a function of the prices p_1 , p_2 , w , the total number of hours H , and the parameters α_1 , α_2 , and γ . [Hint: combine the first two FOCs and then combine the first and third FOCs, and then finally, plug those into the budget constraint to solve for x_1^* . Similarly solve for x_2^* and l^* .]